

Final Exam

Name: _____

Please circle your section number!

SECTION 1

SECTION 2

SECTION 3

SECTION 4

D. KWON
(TuTh 1PM)

BENEDEK VALKÓ
(MWF 11AM)

T. SHCHERBYNA
(TuTh 11AM)

D. CLANCY
(MWF 2:25PM)

- There are 7 problems on the exam, some of them have multiple parts.
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** (Unless it is explicitly stated that this is not required.) In problems 2.-7. numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the attached table on page 3 for the values of the $\Phi(x)$ function.
- On the second page you can find a list of named distributions with their PMF/PDF.
- You can use the empty pages for additional space for writing if necessary.

Problem	Points
1	/18
2	/12
3	/14
4	/14
5	/12
6	/20
7	/10
Total	/100

List of named distributions

$$X \sim \text{Bernoulli}(p), \quad \text{PMF: } p_X(0) = 1 - p, \quad p_X(1) = p$$

$$X \sim \text{Binomial}(n, p), \quad \text{PMF: } p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n$$

$$X \sim \text{Geometric}(p), \quad \text{PMF: } p_X(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

$$X \sim \text{Poisson}(\lambda), \quad \text{PMF: } p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

$$X \sim \text{Hypergeometric}(N, N_A, n), \quad \text{PMF: } p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}, \quad 0 \leq k \leq n$$

$$X \sim \text{Uniform}(a, b), \quad \text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Normal}(\mu, \sigma^2), \quad \text{PDF: } f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$$

$$X \sim \text{Exponential}(\lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Gamma}(r, \lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$(X_1, \dots, X_k) \sim \text{Multinomial}(n, p_1, \dots, p_k),$$

$$\text{joint PMF: } p(a_1, \dots, a_k) = \binom{n}{a_1, \dots, a_k} p_1^{a_1} \cdots p_k^{a_k} \text{ for } a_1, \dots, a_k \geq 0 \text{ with } a_1 + \dots + a_k = n.$$

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

1. **Short problems. (3 points each)** You do not need to justify your answers in this part of the exam.

Suppose that X, Y are discrete random variables that take non-negative integer values with joint probability mass function $p(x, y)$. Express the following quantities in terms of $p(x, y)$.

(Some answers might be in a form incorporating one or more sum. Make sure that the limits of the sums are properly identified. Your expressions should not include other functions besides $p(x, y)$.)

- (a) The probability mass function of X evaluated at 4.

- (b) $P(\max(X, Y) = 1)$

(Problem 1 continued)

(c) $P(X - Y = 1)$

(d) $E[X \cdot e^Y]$

(Problem 1 continued)

(e) The conditional probability mass function of X given $Y = 3$, evaluated at 6.

(f) $E[X|Y = 3]$

2. **(12 points)** We roll a fair die 20 times. Let X be the number of results we obtained exactly 3 times. (So if we get the following twenty numbers in some order: 1,1,1,1,2,2,2,3,3,3,3,3,4,4,4,5,6,6,6,6, then $X = 2$ because of the twos and the fours.) Find the expected value of X .
(You do not need to simplify your answer, but it should be a number, not a sum.)

3. Suppose that X is a non-negative random variable.

(a) **(4 points)** We know that the mean of X is 100. What upper bound can you provide on $P(X \geq 150)$ using this (and only this) information?

(b) **(4 points)** Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 150)$?

(Problem 3 continued)

- (c) **(6 points)** Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 100 i.i.d. random variables. How would you estimate the probability $P(X \geq 150)$ now?

4. Suppose that we have the following information about the random variables X and Y :

$$E[X] = 1, \quad E[X^2] = 10, \quad E[Y] = 2, \quad E[Y^2] = 13, \quad \text{and} \quad \text{Cov}(3X + 2Y, 2Y - 3) = -6.$$

(a) **(3 points)** Find the value of $E[X + 3Y]$.

(b) **(3 points)** Find $\text{Var}(Y)$.

(Problem 5 continued)

(c) **(5 points)** Find the value of $\text{Cov}(X, Y)$.

(d) **(3 points)** Find the correlation of X and Y .

5. **(12 points)** Let $X \sim \text{Exp}(1)$ and $Y \sim \text{Unif}(0, 1)$ be independent. What is the probability density function of the random variable $Z = X + Y$?

Make sure that you give the value of the PDF for all real numbers!

6. X is a continuous random variable with probability density function given by $f_X(x) = 12x^2(1 - x)$ for $0 < x < 1$ and 0 otherwise. The conditional distribution of Y given $X = x$ (for $0 < x < 1$) is uniform on the interval $(x, 1)$.

(a) **(4 points)** Find the conditional probability $P(Y > 0.8|X = 0.6)$.

(b) **(4 points)** Find the conditional expectation $E[Y|X]$.

(Problem 6 continued)

(c) **(4 points)** Find the probability density function of Y .

(d) **(4 points)** Find the conditional probability density function of X given $Y = y$.
(Make sure to specify when the conditional PDF is defined.)

(Problem 6 continued)

- (e) **(4 points)** Find two different single variable integrals for computing $E[Y]$ using the previous parts.
(You do not need to evaluate the integrals.)

7. **(10 points)** You are given a fair die and a fair coin. You roll the die and flip the coin at the same time, and you repeat this once every minute. What is the probability that you roll your first 5 on the die exactly two minutes after seeing tails on the coin for the first time?

(You do not need to simplify your answer, but it should be a number, not a sum.)

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Final Exam – Solutions

1. **Short problems.** You do not need to justify your answers in this part of the exam.

Suppose that X, Y are discrete random variables that take non-negative integer values with joint probability mass function $p(x, y)$. Express the following quantities in terms of $p(x, y)$. (Some answers might be in a form incorporating one or more sum. Make sure that the limits of the sums are properly identified. Your expressions should not include other functions besides $p(x, y)$.)

- (a) The probability mass function of X evaluated at 4.

Solution: $P(X = 4) = \sum_{l=0}^{\infty} P(X = 4, Y = l) = \sum_{l=0}^{\infty} p(4, l).$

- (b) $P(\max(X, Y) = 1)$

Solution: Since X and Y take only non-negative integer values, $\max(X, Y) = 1$ only if both X and Y are equal 1 or if one is 1 the other is 0. Thus

$$P(\max(X, Y) = 1) = p(0, 1) + p(1, 0) + p(1, 1).$$

- (c) $P(X - Y = 1)$

Solution: Since X and Y take only non-negative integer values, $X - Y = 1$ if and only if $X = k + 1$, $Y = k$ for some non-negative integer $k \geq 0$. Thus

$$P(X - Y = 1) = \sum_{k=0}^{\infty} p(k + 1, k).$$

- (d) $E[X \cdot e^Y]$

Solution: $E[X \cdot e^Y] = \sum_{k,l=0}^{\infty} k \cdot e^l \cdot p(k, l).$

- (e) The conditional probability mass function of X given $Y = 3$, evaluated at 6.

Solution: $p_{X|Y}(6|3) = \frac{p(6,3)}{P(Y=3)} = \frac{p(6,3)}{\sum_{k=0}^{\infty} p(k,3)}.$

If $P(Y = 3) = \sum_{k=0}^{\infty} p(k, 3) = 0$ then this is not defined.

- (f) $E[X|Y = 3]$ **Solution:** By definition,

$$E[X|Y = 3] = \sum_{k=0}^{\infty} k \cdot p_{X|Y}(k|3).$$

Substituting $p_{X|Y}(k|3)$ from (e), we get

$$E[X|Y = 3] = \frac{\sum_{\ell=0}^{\infty} \ell p(\ell, 3)}{\sum_{k=0}^{\infty} p(k, 3)}$$

If $P(Y = 3) = \sum_{k=0}^{\infty} p(k, 3) = 0$ then this is not defined.

2. We roll a fair die 20 times. Let X be the number of results we obtained exactly 3 times. (So if we get the following twenty numbers in some order: 1,1,1,1,2,2,2,3,3,3,3,3,4,4,4,5,6,6,6,6, then $X = 2$ because of the twos and the fours.) Find the expected value of X .

(You do not need to simplify your answer, but it should be a number, not a sum.)

Solution: Define random variables I_k , $k = 1, 2, \dots, 6$ to be 1 if the result k was obtained exactly three times, and 0 otherwise. Then

$$X = I_1 + I_2 + \dots + I_6,$$

and by linearity and exchangeability

$$E[X] = E[I_1] + E[I_2] + \dots + E[I_6] = 6E[I_1].$$

Now

$$E[I_1] = P(1 \text{ was obtained exactly 3 times}) = \binom{20}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{17},$$

thus

$$E[X] = 6 \cdot \binom{20}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{17}.$$

3. Suppose that X is a non-negative random variable.

- (a) We know that the mean of X is 100. How can you estimate $P(X \geq 150)$ using this (and only this) information?

Solution: Since X is non-negative, according to Markov's inequality, we have

$$P(X \geq 150) \leq \frac{E[X]}{150} = \frac{100}{150} = \frac{2}{3}.$$

- (b) Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 150)$?

Solution: According to Chebyshev's inequality, we get

$$P(X \geq 150) = P(X - 100 \geq 50) \leq P(|X - 100| \geq 50) \leq \frac{\text{Var}(X)}{50^2} = \frac{400}{2500} = \frac{4}{25}.$$

- (c) Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 100 i.i.d. random variables. How would you estimate the probability $P(X \geq 150)$ now?

Solution: We can apply CLT here. Since $\sqrt{\text{Var}(X)} = 20$, we get

$$P(X \geq 150) = P\left(\frac{X - 100}{20} \geq \frac{5}{2}\right) \approx P(Z \geq 5/2) = 1 - \Phi(2.5) = 1 - 0.9938 = 0.0062.$$

Here Z is the standard normal random variable.

Side note: if we denote the 100 iid random variables by $Y_1, \dots, Y_{100}$ then $E[Y_1] = \frac{E[X]}{100} = 1$ and $\text{Var}(Y_1) = \frac{\text{Var}(X)}{100} = 4$, but this is actually not required for the solution.

4. Suppose that we have the following information about the random variables X and Y :

$$E[X] = 1, \quad E[X^2] = 10, \quad E[Y] = 2, \quad E[Y^2] = 13, \quad \text{and} \quad \text{Cov}(3X + 2Y, 2Y - 3) = -6.$$

(a) Find the value of $E[X + 3Y]$.

Solution: Because of the linearity of expectation, we get

$$E[X + 3Y] = E[X] + 3E[Y] = 1 + 3 \cdot 2 = 7.$$

(b) Find $\text{Var}(Y)$.

$$\text{Solution: } \text{Var}(Y) = E[Y^2] - (E[Y])^2 = 13 - 4 = 9.$$

(c) What is $\text{Cov}(X, Y)$?

Solution: Since

$$\text{Cov}(3X + 2Y, 2Y - 3) = \text{Cov}(3X, 2Y) + \text{Cov}(2Y, 2Y) = 6\text{Cov}(X, Y) + 4\text{Var}(Y),$$

we can use (b) to get

$$\text{Cov}(X, Y) = \frac{-6 - 4\text{Var}(Y)}{6} = -\frac{42}{6} = -7.$$

(d) What is $\text{Corr}(X, Y)$?

Solution: Since

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 10 - 1 = 9,$$

we get from (b) and (c)

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = -\frac{7}{9}.$$

5. Let $X \sim \text{Exp}(1)$ and $Y \sim \text{Unif}(0, 1)$ be independent. What is the probability density function of the random variable $Z = X + Y$?

Make sure that you give the value of the PDF for all real numbers!

Solution: Because of independence, the PDF of $Z = X + Y$ is given by the convolution of PDF of X and PDF of Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x) dx.$$

$f_X(x) \neq 0$ only if $x > 0$, and $f_Y(z - x) \neq 0$ only if $z - x \in (0, 1)$ or, equivalently, $x \in (z - 1, z)$. Thus,

$$f_Z(z) = 0,$$

if $z \leq 0$; if $z \geq 1$ we get

$$f_Z(z) = \int_{z-1}^z e^{-x} dx = e^{-(z-1)} - e^{-z},$$

and if $z \in (0, 1)$, we have

$$f_Z(z) = \int_0^z e^{-x} dx = 1 - e^{-z}.$$

So finally

$$f_Z(z) = \begin{cases} e^{-(z-1)} - e^{-z}, & z \geq 1, \\ 1 - e^{-z}, & z \in [0, 1), \\ 0, & z < 0. \end{cases}$$

6. X is a continuous random variable with probability density function given by $f_X(x) = 12x^2(1-x)$ for $x \in (0, 1)$ and 0 otherwise. The conditional distribution of Y given $X = x$ (for $0 < x < 1$) is uniform on the interval $(x, 1)$.

- (a) **(4 points)** Find the conditional probability $P(Y > 0.8|X = 0.6)$.

Solution: The conditional distribution of Y given $X = 0.6$ is uniform on $(0.6, 1)$ so the conditional probability of being larger than 0.8 is $\frac{1-0.8}{1-0.6} = \frac{1}{2}$.

- (b) **(4 points)** Find the conditional expectation $E[Y|X]$.

Solution: The conditional distribution of Y given $X = x$ is uniform on $(x, 1)$ so the conditional expectation given $X = x$ (for $0 < x < 1$) is $\frac{x+1}{2}$ (from the expected value of the uniform distribution). Hence

$$E[Y|X = x] = \frac{1+x}{2},$$

and replacing x with X gives

$$E[Y|X] = \frac{1+X}{2}.$$

- (c) **(4 points)** Find the probability density function of Y .

Solution: Note that X values from $(0, 1)$ and Y is between X and 1, so (X, Y) takes values from $\{(x, y) : 0 < x < y < 1\}$. The conditional density of Y given $X = x$ (for $0 < x < 1$) is the PDF of the uniform density on $(x, 1)$ which is $\frac{1}{1-x}$ on $(x, 1)$ and 0 outside. Hence the joint density of (X, Y) (for $0 < x < y < 1$) is

$$f(x, y) = f_{Y|X}(y|x)f_X(x) = \frac{1}{1-x}12x^2(1-x) = 12x^2,$$

with $f(x, y) = 0$ for all other values.

$$f(x, y) = \begin{cases} 12x^2, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

We can find the PDF of Y by integrating out x from the joint density. Assuming $0 < y < 1$:

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_0^y 12x^2dx = 4y^3,$$

and $f_Y(y) = 0$ otherwise.

$$f_Y(y) = \begin{cases} 4y^3, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- (d) **(4 points)** Find the conditional probability density function of X given $Y = y$. (Make sure to specify when the conditional PDF is defined.)

Solution: This is given by $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$. Since we need $f_Y(y) > 0$, this is only defined for $0 < y < 1$. Then we get

$$f_{X|Y}(x|y) = \begin{cases} \frac{12x^2}{4y^3} = 3\frac{x^2}{y^3}, & 0 < x < y \\ 0, & \text{otherwise} \end{cases}$$

- (e) **(4 points)** Find two different single variable integrals for computing $E[Y]$ using the previous parts. (You do not need to evaluate the integrals.)

Solution: Using the PDF:

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 y \cdot 4y^3 dy.$$

Using the averaging principle and $E[Y|X = x]$:

$$E[Y] = \int_{-\infty}^{\infty} E[Y|X = x] f_X(x) dx = \int_0^1 \frac{1+x}{2} 12x^2(1-x) dx.$$

7. **(10 points)** You are given a fair die and a fair coin. You roll the die and flip the coin at the same time, and you repeat this once every minute. What is the probability that you roll your first 5 on the die exactly two minutes after seeing tails on the coin for the first time?

(You do not need to simplify your answer, but it should be a number, not a sum.)

Solution: Let A be the event in question, and denote by A_k the event that the first tail came on the k th flip of the coin and the first 5 came on the $(k+2)$ nd roll of the die. Using the independence of the die rolls and the coin flips we get

$$P(A_k) = \frac{1}{2^{k-1}} \frac{1}{2} \left(\frac{5}{6}\right)^{k+1} \frac{1}{6} = \left(\frac{5}{12}\right)^k \frac{5}{36}$$

Since A is the disjoint union of the events $A_1, A_2, \dots$ we get

$$P(A) = \sum_{k=1}^{\infty} \left(\frac{5}{12}\right)^k \frac{5}{36} = \frac{5}{36} \frac{5/12}{1 - 5/12} = \frac{25}{7 \cdot 36} = \frac{25}{252}$$

Final Exam

Name: _____

Please circle your section number!

SECTION 1	SECTION 2	SECTION 3	SECTION 4
AMY COCHRAN (MWF 9:55p)	BENEDEK VALKÓ (MWF 12:05p)	SARAH TAMMEN (TuTh 9:30a)	SHAOMING GUO (MWF 8:50a)

- **There are 6 problems on the exam, some of them have multiple parts.** The problems are on pages 3 to 14. Make sure to read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.** You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Your answer can be left in terms of the cumulative distribution function Φ of a standard normal random variable or in terms of the exponential function e^x , e.g. expressions like $\Phi(1)$ or $3e^{-9}$ would not need to be simplified further.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	20	
2	15	
3	20	
4	19	
5	10	
6	16	
Total:	100	

List of named distributions

$$X \sim \text{Bernoulli}(p), \quad \text{PMF: } p_X(0) = 1 - p, \quad p_X(1) = p$$

$$X \sim \text{Binomial}(n, p), \quad \text{PMF: } p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n$$

$$X \sim \text{Geometric}(p), \quad \text{PMF: } p_X(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

$$X \sim \text{Poisson}(\lambda), \quad \text{PMF: } p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

$$X \sim \text{Hypergeometric}(N, N_A, n), \quad \text{PMF: } p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}, \quad 0 \leq k \leq n$$

$$X \sim \text{Uniform}(a, b), \quad \text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Normal}(\mu, \sigma^2), \quad \text{PDF: } f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}, \quad t \in \mathbb{R}$$

$$X \sim \text{Exponential}(\lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Gamma}(r, \lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

Here $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$, and $\Gamma(n) = (n-1)!$ for $n \geq 1$ integer.

$$(X_1, \dots, X_k) \sim \text{Multinomial}(n, k, p_1, \dots, p_k),$$

$$\text{joint PMF: } p(a_1, \dots, a_k) = \frac{n!}{a_1! a_2! \dots a_k!} p_1^{a_1} \dots p_k^{a_k} \text{ for } a_1, \dots, a_k \geq 0 \text{ with } a_1 + \dots + a_k = n.$$

1. For each of the following problems, we suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables with mean 0 and variance 1.

(a) (5 points) What is the expected value of the random variable $Y = X_1 + 2X_2 + 3X_3X_4$?

(b) (5 points) Find $\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1)$.

(Problem 1 continued)

- (c) (5 points) Suppose that we know that the moment generating function of X_1 is given by $M(t) = \frac{1}{1-t^2/2}$ whenever $|t| < \sqrt{2}$. What is the moment generating function of the random variable $Z = X_1 + X_2$ at $t = 1$?

- (d) (5 points) Suppose that X_1 has standard normal distribution. Find $P(W > 0)$ for $W = X_1 + 2X_2 - X_3$.

2. Jrue is a professional basketball player, averaging 16 points per game. Let X denote the number of points he scores in his next basketball game.

Note. You will estimate and/or approximate certain probabilities in some of the following problems. Justify the approach you use!

- (a) (5 points) What is the best upper bound you can provide for $P(X \geq 30)$?

- (b) (5 points) Suppose that we also learn that the variance of X is approximately 36. How can you improve on your upper bound on $P(X \geq 30)$?

(Problem 2 continued)

- (c) (5 points) Approximate the probability that in the next season (100 games, counting playoffs), Jrue will score more than 1700 points in total.

Note. You may assume that the number of points scored by Jrue in different games are independent and identically distributed.

3. We flip a fair coin. If it is heads, we choose a number uniformly at random from $[0, 1]$. If it is tails, we choose a number uniformly at random from $[0, 2]$. Let X denote the chosen number.

(a) (5 points) Find the probability that $X \leq 1/2$.

(b) (5 points) Find the probability that we rolled heads given that $X \leq 1/2$.

(Problem 3 continued)

- (c) (5 points) Find the cumulative distribution function of X . (Make sure that the function you provide is defined on the whole line.)

(Problem 3 continued)

(d) (5 points) Show that X is continuous, and find its probability density function.

4. Suppose we have jointly continuous X and Y with joint probability density function:

$$f(x, y) = \begin{cases} \frac{4}{y}e^{-4y} & \text{if } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) (6 points) Let $f_Y(y)$ denote the probability density function of Y . Find $f_Y(y)$ and identify the distribution of Y by name.

Note. Make sure to identify the parameters of the distribution.

- (b) (6 points) Let $f_{X|Y}(x|y)$ denote the conditional density function of X given $Y = y$. Find $f_{X|Y}(x|y)$ and identify the conditional distribution of X given $Y = y$ by name.

Note. Make sure to identify the parameters of the distribution.

(Problem 4 continued)

(c) (3 points) Find $E[X|Y]$.

(d) (4 points) Find $E[X]$.

5. Suppose that X and Y are independent continuous random variables with probability density functions f_X and f_Y .

(a) (3 points) Write down the integral formula for the probability density function of $Z = X + Y$.

(b) (7 points) Suppose that $X \sim N(0, 1)$ and $Y \sim \text{Unif}[0, 1]$. Find the probability density function of $Z = X + Y$ at $z = 7$ using the integral formula given in (a).

Note. Your final answer should be in terms of the Φ function.

6. A monkey is sitting in front of a keyboard that has the 26 capital letters of the alphabet (and nothing else). He starts randomly hitting the keys: he chooses the keys independently and uniformly.
- (a) (4 points) What is the probability that the first two keystrokes are U and W in this order?
- (b) (8 points) We print out the first 1000 keystrokes (in order). Let X be the number of times we see the sequence UW spelled out (these 2 letters next to each other) somewhere in that text. Find $E[X]$.

(Problem 6 continued)

(c) (4 points) Find $\text{Var}(X)$ for the random variable X in part (b).

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Final Exam - Solution outlines

1. For each of the following problems, we suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables with mean 0 and variance 1.

- (a) (5 points) What is the expected value of the random variable $Y = X_1 + 2X_2 + 3X_3X_4$?

Solution: Using the linearity of expectation we get

$$E[X_1 + 2X_2 + 3X_3X_4] = E[X_1] + 2E[X_2] + 3E[X_3X_4]$$

Since X_3 and X_4 are independent, we also have $E[X_3X_4] = E[X_3]E[X_4]$. Since the random variables are mean zero, we get

$$E[X_1 + 2X_2 + 3X_3X_4] = E[X_1] + 2E[X_2] + 3E[X_3]E[X_4] = 0$$

- (b) (5 points) Find $\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1)$.

Solution: The covariance of two random variables does not change if we add a constant to one of them, hence

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1) = \text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3).$$

Using $\text{Cov}(Y_1 + Y_2, Z) = \text{Cov}(Y_1, Z) + \text{Cov}(Y_2, Z)$ and $\text{Cov}(Y, Z) = \text{Cov}(Z, Y)$ repeatedly we get

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = \text{Cov}(X_1, 3X_2) + \text{Cov}(X_1, 4X_3) + \text{Cov}(2X_2, 3X_2) + \text{Cov}(2X_2, 4X_3)$$

Using $\text{Cov}(aY, Z) = a \text{Cov}(Y, Z)$ leads to

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = 3 \text{Cov}(X_1, X_2) + 4 \text{Cov}(X_1, X_3) + 6 \text{Cov}(X_2, X_2) + 8 \text{Cov}(X_2, X_3).$$

Independent random variables have covariance 0, and since X_1, X_2, X_3, X_4 are independent, we get

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = 6 \text{Cov}(X_2, X_2) = 6 \text{Var}(X_2) = 6.$$

In the last step we used that the covariance of a random variable with itself is the variance.

- (c) (5 points) Suppose that we know that the moment generating function of X_1 is given by $M(t) = \frac{1}{1-t^2/2}$ whenever $|t| < \sqrt{2}$. What is the moment generating function of the random variable $Z = X_1 + X_2$ at $t = 1$?

Solution: Since X_1 and X_2 are independent, the MGF of their sum is just the product of the individual MGFs:

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t).$$

Since X_1 and X_2 have identical distributions, their MGFs are the same (and equal to $M(t)$):

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = M(t)^2$$

Plugging in $t = 1$ gives

$$M_{X_1+X_2}(t) = M(t)^2 = \frac{1}{(1-1/2)^2} = 4.$$

- (d) (5 points) Suppose that X_1 has standard normal distribution. Find $P(W > 0)$ for $W = X_1 + 2X_2 - X_3$.

Solution: We have seen in class that a linear combination of independent normals is also a normal. Hence W has normal distribution, the mean is equal to $E[X_1 + 2X_2 - X_3] = E[X_1] + 2E[X_2] - E[X_3] = 0$, and the variance is

$$\text{Var}(X_1 + 2X_2 - X_3) = \text{Var}(X_1) + 2^2 \text{Var}(X_2) + (-1)^2 \text{Var}(X_3) = 6.$$

(We used in the last step that the variance of the sum of independent random variables is the sum of the variances.) Hence

$$P(W > 0) = P(Z > 0)$$

where $Z \sim N(0, 6)$. Then $Z/\sqrt{6}$ is a standard normal and we get

$$P(Z > 0) = P\left(Z/\sqrt{6} > 0\right) = 1 - \Phi(0) = 1/2.$$

2. Jrue is a professional basketball player, averaging 16 points per game. Let X denote the number of points he scores in his next basketball game.

Note. You will estimate and/or approximate certain probabilities in some of the following problems. Justify the approach you use!

- (a) (5 points) What is the best upper bound you can provide for $P(X \geq 30)$?

Solution: Since X is a non-negative random variable, we can apply Markov's inequality:

$$P(X \geq 30) \leq \frac{E[X]}{30} = \frac{16}{30} = \frac{8}{15}.$$

- (b) (5 points) Suppose that we also learn that the variance of X is approximately 36. How can you improve on your upper bound on $P(X \geq 30)$?

Solution: We can use Chebyshev's inequality to estimate this probability. For this we first rewrite the event as an inequality on the deviation from the mean:

$$\begin{aligned} P(X \geq 30) &= P(X - E[X] \geq 30 - E[X]) = P(X - E[X] \geq 14) \leq P(|X - E[X]| \geq 14) \\ &\leq \frac{\text{Var}(X)}{14^2} = \frac{36}{14^2} = \frac{9}{49}. \end{aligned}$$

- (c) (5 points) Approximate the probability that in the next season (100 games, counting playoffs), Jrue will score more than 1700 points in total.

Note. You may assume that the number of points scored by Jrue in different games are independent and identically distributed.

Solution: Let $X_1, \dots, X_{100}$ denote the points scored in the various games next season, and let S denote their sum. Since $X_1, \dots, X_{100}$ are iid by assumption we may use the Central Limit Theorem to approximate the standardized version of S with a standard normal random variable Z . Using the fact that the random variables are iid we get

$$E[S] = 100E[X_1] = 1600, \quad \text{Var}(S) = 100 \text{Var}(X_1) = 3600 = 60^2,$$

so the standardized version of S is $\frac{S - E[S]}{\sqrt{\text{Var}(S)}} = \frac{S - 1600}{60}$. This leads to

$$P(S > 1700) = P\left(\frac{S - 1600}{60} > \frac{1700 - 1600}{60}\right) \approx P\left(Z > \frac{5}{3}\right) = 1 - \Phi(5/3).$$

3. We flip a fair coin. If it is heads, we choose a number uniformly at random from $[0, 1]$. If it is tails, we choose a number uniformly at random from $[0, 2]$. Let X denote the chosen number.

(a) (5 points) Find the probability that $X \leq 1/2$.

Solution: We will condition on the outcome of the coin flip. Let B be the event that the coin lands heads, and let $A = \{X \leq 1/2\}$. Then by the law of total probability we have

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c).$$

We have $P(B) = P(B^c) = 1/2$. From the setup we know that

$$P(A|B) = P(X < 1/2|B) = \frac{1/2}{1} = 1/2, \quad P(A|B^c) = P(X < 1/2|B^c) = \frac{1/2}{2} = 1/4.$$

This gives

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

(b) (5 points) Find the probability that we rolled heads given that $X \leq 1/2$.

Solution: Using the notation of part (a) we need $P(B|A)$, which we can compute using Bayes formula and the probabilities from part (a):

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)} = \frac{1/2 \cdot 1/2}{3/8} = \frac{2}{3}.$$

(c) (5 points) Find the cumulative distribution function of X . (Make sure that the function you provide is defined on the whole line.)

Solution: We need to compute $F(t) = P(X \leq t)$ for all values of t . Note that X can only take values from $[0, 2]$, hence $F(t) = 0$ for $t < 0$ and $F(t) = 1$ for $t \geq 2$.

For $t \in [0, 2)$ we can compute $P(X \leq t)$ by conditioning, just as in part (a).

$$P(X \leq t) = P(X \leq t|B)P(B) + P(X \leq t|B^c)P(B^c)$$

If we flip heads the conditional distribution of X is uniform on $[0, 1]$, hence $P(X \leq t|B)$ is equal to t for $t \in [0, 1]$ and equal to 1 if $t > 1$.

If we flip tails the conditional distribution of X is uniform on $[0, 2]$, hence for $t \in [0, 2)$ we have $P(X \leq t|B^c) = \frac{t}{2}$.

Hence for $t \in [0, 1]$ we have

$$F(t) = t \cdot 1/2 + \frac{t}{2} \cdot 1/2 = \frac{3}{4}t,$$

and for $t \in [1, 2)$ we have

$$F(t) = 1 \cdot 1/2 + \frac{t}{2} \cdot 1/2 = \frac{t+2}{4}.$$

This leads to

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{3}{4}t, & 0 \leq t \leq 1 \\ \frac{t+2}{4}, & 1 < t \leq 2 \\ 1, & 2 \leq t. \end{cases}$$

- (d) (5 points) Show that X is continuous, and find its probability density function.

Solution: The CDF of X is continuous everywhere, and it is differentiable apart from finitely many values. Hence X has continuous distribution, and the PDF can be computed by differentiating F :

$$f_X(t) = \begin{cases} \frac{3}{4}, & 0 \leq t \leq 1 \\ \frac{1}{4}, & 1 < t \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

4. Suppose we have jointly continuous X and Y with joint probability density function:

$$f(x, y) = \begin{cases} \frac{4}{y}e^{-4y} & \text{if } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) (6 points) Let $f_Y(y)$ denote the probability density function of Y . Find $f_Y(y)$ and identify the distribution of Y by name.

Note. Make sure to identify the parameters of the distribution.

Solution: The marginal PDF is computed as $f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx$. Note that $f(x, y) = 0$ for $y \leq 0$, so we get $f_Y(y) = 0$ in this case. For $y > 0$ we have

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_0^y \frac{4}{y}e^{-4y}dx = 4e^{-4y}$$

(note that we are integrating a constant). The resulting PDF is exactly the PDF of the $\text{Exp}(4)$ distribution.

- (b) (6 points) Let $f_{X|Y}(x|y)$ denote the conditional density function of X given $Y = y$. Find $f_{X|Y}(x|y)$ and identify the conditional distribution of X given $Y = y$ by name.

Note. Make sure to identify the parameters of the distribution.

Solution: For jointly continuous random variables X, Y the conditional PDF $f_{X|Y}(x|y)$ is defined if $f_Y(y) > 0$, and in that case $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$. Hence, for $y > 0$ we have

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{\frac{4}{y}e^{-4y}I(0 < x < y)}{4e^{-4y}} = \frac{1}{y}I(0 < x < y).$$

Here $I(0 < x < y)$ is the indicator of $0 < x < y$: it is 1 if $x \in (0, y)$ and 0 otherwise. (One can also write this as a case-defined function.) The resulting conditional PDF is exactly the PDF of the $\text{Uniform}(0, y)$ distribution.

- (c) (3 points) Find $E[X|Y]$.

Solution: We first compute $E[X|Y = y]$. This is well-defined if $f_Y(y) > 0$, which happens if $y > 0$. Since the conditional distribution of X is $\text{Uniform}(0, y)$, and the expectation of the uniform is the midpoint of the interval, we get $E[X|Y = y] = \frac{y}{2}$. From this we get $E[X|Y] = \frac{Y}{2}$.

- (d) (4 points) Find $E[X]$.

Solution: We can compute $E[X]$ by conditioning on Y first and taking the expected value of $E[X|Y]$:

$$E[X] = E[E[X|Y]] = E[Y/2] = \frac{1}{2}E[Y].$$

We have seen that $Y \sim \text{Exp}(4)$, and we know that the expectation of the exponential distribution is one over the parameter, hence

$$E[X] = \frac{1}{2}E[Y] = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

One can also compute this by evaluating the double integral

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy = \int_0^{\infty} \int_0^y x \frac{4}{y} e^{-4y} dx dy.$$

5. Suppose that X and Y are independent continuous random variables with probability density functions f_X and f_Y .

- (a) (3 points) Write down the integral formula for the probability density function of $Z = X + Y$.

Solution: Because X and Y are independent and continuous, we can use the convolution formula given by

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

- (b) (7 points) Suppose that $X \sim N(0, 1)$ and $Y \sim \text{Unif}[0, 1]$. Find the probability density function of $Z = X + Y$ at $z = 7$ using the integral formula given in (a).

Note. Your final answer should be in terms of the Φ function.

Solution: By the convolution formula

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(7 - x) dx.$$

We have $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $f_Y(y) = I(0 < y < 1)$. Hence $f_X(x) f_Y(7 - x)$ is positive exactly if $0 < 7 - x < 1$, which is equivalent to $6 < x < 7$. In that case the $f_X(x) f_Y(7 - x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, so

$$f_{X+Y}(7) = \int_{-\infty}^{\infty} f_X(x) f_Y(7 - x) dx = \int_6^7 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

This is the integral of the PDF of the standard normal on $[6, 7]$, which is just the probability of the standard normal being in that interval. We can compute this with the CDF of the standard normal to get $\Phi(7) - \Phi(6)$. Hence

$$f_{X+Y}(7) = \Phi(7) - \Phi(6).$$

6. A monkey is sitting in front of a keyboard that has the 26 capital letters of the alphabet (and nothing else). He starts randomly hitting the keys: he chooses the keys independently and uniformly.

- (a) (4 points) What is the probability that the first two keystrokes are U and W in this order?

Solution: Let A be the event that the first keystroke is U, B the event that the second keystroke is W. By our assumption $P(A) = P(B) = \frac{1}{26}$, and the two events are independent, hence

$$P(\text{first two letters are U and W in this order}) = P(AB) = \frac{1}{26^2}$$

We can also compute this probability by looking at the first two letters, and note that we have 26^2 equally likely outcomes.

- (b) (8 points) We print out the first 1000 keystrokes (in order). Let X be the number of times we see the sequence UW spelled out (these 2 letters next to each other) somewhere in that text. Find $E[X]$.

Solution: Let I_j be the indicator of the event that the j th letter is U and the $(j + 1)$ st letter is W. Then

$$X = I_1 + \cdots + I_{999}.$$

By linearity

$$E[X] = \sum_{j=1}^{999} E[I_j].$$

We have

$$\begin{aligned} E[I_j] &= P(j\text{th letter is U and the } (j + 1)\text{st letter is W}) \\ &= P(1\text{st letter is U and the 2nd letter is W}) \\ &= \frac{1}{26^2} \end{aligned}$$

where we used the fact that the letters are iid, and hence are exchangeable.

This gives $E[X] = \frac{999}{26^2}$.

- (c) (4 points) Find $\text{Var}(X)$ for the random variable X in part (b).

Solution: Using the notation of part (b) and the formula for the variance of the sum of random variables

$$\text{Var}(X) = \text{Var}\left(\sum_{j=1}^{999} I_j\right) = \sum_{j=1}^{999} \text{Var}(I_j) + 2 \sum_{1 \leq i < j \leq 999} \text{Cov}(I_i, I_j).$$

We have $\text{Var}(I_j) = p(1 - p)$ where $p = \frac{1}{26^2}$ is the probability of $I_j = 1$, as computed in part (a).

Note that I_i depends on the i th and $(i + 1)$ st letter, and I_j depends on the j th and $(j + 1)$ st letters. If $i + 1 < j$ then these letters are all different, and hence are independent, which means that in this case $\text{Cov}(I_i, I_j) = 0$.

If $i + 1 = j$ then we need to compute

$$\text{Cov}(I_i, I_{i+1}) = E[I_i I_{i+1}] - E[I_i]E[I_{i+1}].$$

Not that $I_i I_{i+1}$ can only be nonzero if $I_i = I_{i+1} = 1$, which cannot happen. (If $I_i = 1$ then the $(i + 1)$ st letter is W, if $I_{i+1} = 1$ then the $(i + 1)$ st letter is U.) Hence $E[I_i I_{i+1}] = 0$ and

$$\text{Cov}(I_i, I_{i+1}) = E[I_i I_{i+1}] - E[I_i]E[I_{i+1}] = -\frac{1}{26^4}$$

using part (a). Putting everything together

$$\begin{aligned} \text{Var}(X) &= \text{Var}\left(\sum_{j=1}^{999} I_j\right) = \sum_{j=1}^{999} \text{Var}(I_j) + 2 \sum_{1 \leq i < j \leq 999} \text{Cov}(I_i, I_j) \\ &= 999 \text{Var}(I_1) + 2 \sum_{i=1}^{998} \text{Cov}(I_i, I_{i+1}) = 999 \frac{1}{26^2} \left(1 - \frac{1}{26^2}\right) - 2 \cdot 998 \frac{1}{26^4} \end{aligned}$$

Department of Mathematics, University of Wisconsin-Madison
Math 431 — Final Exam — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

Time: **120 minutes**

- This exam contains 7 questions some with multiple parts, 12 pages (including the cover) for the total of 91 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
- Justify your steps.
- For binomial random variable approximations, you will NOT get full credit for your answer if you do not include some explanation of why that approximation is valid.
- You can use the attached table of the values of the $\Phi(x)$ function.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	7	Total
Points:	16	16	11	12	8	20	8	91
Score:								

First Name: _____

Last Name: _____

1. For this problem, you need to provide **answers only**.

(a) (4 points) Find $\text{Corr}(-2X + 1, X)$.

(b) (4 points) Suppose A, B, C are events on one probability space. Write down explicit conditions for them to be mutually independent.

(c) (4 points) Let X and Y be random variables such that $X \sim N(0, 1)$ and conditional on $X = x$, Y is $N(x, 1)$. Find the PDF of Y as an integral.

(d) (4 points) X_1 and X_2 are iid random variables with the cumulative distribution function $F(t)$. Find the cumulative distribution function of $Y = \max(X_1, X_2)$.

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2. Suppose that X is a non-negative random variable.

(a) (5 points) We know that the mean of X is 80. How can you estimate $P(X \geq 120)$ using this (and only this) information?

(b) (5 points) Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 120)$?

(c) (6 points) Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 200 i.i.d. random variables. How would you estimate the probability $P(X \geq 120)$ now?

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3. (a) (5 points) A fair 6-sided die is rolled once. Find the expected number of additional rolls needed to obtain a value at least as large as that of the first roll, given that the first roll is 3.

- (b) (6 points) A fair 6-sided die is rolled once. Find the expected number of additional rolls needed to obtain a value at least as large as that of the first roll.

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4. We are given a biased coin and we are told that because of manufacturing defects, the probability of heads, denoted by Y , is itself random, with a distribution over the interval $[0, 0.5]$ given by density $1 + 4y$. We toss the coin a fixed number n of times, and we let X be the number of heads obtained.

(a) (6 points) Find $E[X|Y]$.

(b) (6 points) Find $E[X]$.

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5. (8 points) Suppose $X \sim \text{Exp}(\lambda)$, that is, X is an exponential random variable with parameter $\lambda > 0$. Find the probability density function of X^2 .

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6. An urn contains six balls, one is red, three are green, two are yellow. Randomly draw three balls from the urn **without** replacement, and let R and G denote the number of red and green balls (within the three balls), respectively.
- (a) (6 points) Compute $P(R = 1, G = 1)$.

- (b) (6 points) Compute the expectations $E[R]$ and $E[G]$.

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(c) (8 points) Compute the covariance $\text{Cov}(R, G)$.

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7. (8 points) Let X and Y be independent random variables with distributions $\text{Unif}(0, 2)$ and $\text{Unif}(1, 3)$. Let $W = X + Y$. Find the probability density function of W .

Table of Named Distributions

Name	p.m.f. or p.d.f.	$E[X]$	$Var(X)$
Ber(p)	$p_X(0) = 1 - p, p_X(1) = p$	p	$p(1 - p)$
Bin(n, p)	$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, 0 \leq k \leq n$	np	$np(1 - p)$
Geom(p)	$p_X(k) = p(1 - p)^{k-1}, k \geq 1$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
NegBin(k, p)	$p_X(n) = \binom{n-1}{k-1} p^k (1 - p)^{n-k}, n \geq k$	$\frac{k}{p}$	$\frac{k(1 - p)}{p^2}$
HyperGeom(N, N_A, n)	$p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}$	$\frac{nN_A}{N}$	$\frac{nN_A(N - N_A)(N - n)}{N^2(N - 1)}$
Poisson(λ)	$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, k \geq 0$	λ	λ
Unif($[a, b]$)	$f_X(x) = \frac{1}{b - a}, x \in [a, b]$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
N(μ, σ^2)	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$	μ	σ^2
Exp(λ)	$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(n, λ)	$f_X(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n - 1)!}, x \geq 0$	$\frac{n}{\lambda}$	$\frac{n}{\lambda^2}$

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

Department of Mathematics, University of Wisconsin-Madison
Math 431 — Final Exam — Solutions — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

Time: **120 minutes**

- This exam contains 7 questions some with multiple parts, 12 pages (including the cover) for the total of 91 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
- Justify your steps.
- For binomial random variable approximations, you will NOT get full credit for your answer if you do not include some explanation of why that approximation is valid.
- You can use the attached table of the values of the $\Phi(x)$ function.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	7	Total
Points:	16	16	11	12	8	20	8	91
Score:								

1. For this problem, you need to provide **answers only**.

(a) (4 points) Find $\text{Corr}(-2X + 1, X)$.

Solution: $\text{Corr}(-2X + 1, X) = \text{Corr}(-2X, X) + \text{Corr}(1, X) = -\text{Corr}(X, X) = \boxed{-1}$.

(b) (4 points) Suppose A, B, C are events on one probability space. Write down explicit conditions for them to be mutually independent.

Solution:

$$\begin{aligned}P(AB) &= P(A)P(B) \\P(AC) &= P(A)P(C) \\P(BC) &= P(B)P(C) \\P(ABC) &= P(A)P(B)P(C)\end{aligned}$$

(c) (4 points) Let X and Y be random variables such that $X \sim N(0, 1)$ and conditional on $X = x$, Y is $N(x, 1)$. Find the PDF of Y as an integral.

Solution:

$$\int_{-\infty}^{\infty} f_{Y|X}(y|x)F_X(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-x)^2} \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}dx$$

(d) (4 points) X_1 and X_2 are iid random variables with the cumulative distribution function $F(t)$. Find the cumulative distribution function of $Y = \max(X_1, X_2)$.

Solution: $F_Y(t) = P(Y \leq t) = P(\max(X_1, X_2) \leq t) = P(X_1 \leq t, X_2 \leq t) = P(X_1 \leq t)P(X_2 \leq t) = \boxed{F(t)^2}$.

2. Suppose that X is a non-negative random variable.

- (a) (5 points) We know that the mean of X is 80. How can you estimate $P(X \geq 120)$ using this (and only this) information?

Solution: Since X is non-negative, according to Markov's inequality, we have

$$P(X \geq 120) \leq \frac{E[X]}{120} = \frac{80}{120} = \frac{2}{3}.$$

- (b) (5 points) Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 120)$?

Solution: According to Chebyshev's inequality, we get

$$P(X \geq 120) = P(X - 80 \geq 40) \leq P(|X - 80| \geq 40) \leq \frac{\text{Var}(X)}{40^2} = \frac{400}{40 \times 40} = \frac{1}{4}.$$

- (c) (6 points) Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 200 i.i.d. random variables. How would you estimate the probability $P(X \geq 120)$ now?

Solution: We can apply CLT here. Since $\sqrt{\text{Var}(X)} = 20$, we get

$$P(X \geq 120) = P\left(\frac{X - 80}{20} \geq \frac{120 - 80}{20}\right) \approx P(Z \geq 2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228.$$

Here Z is the standard normal random variable.

Side note: if we denote the 200 i.i.d. random variables by $Y_1, \dots, Y_{200}$ then $E[Y_1] = \frac{E[X]}{200} = \frac{2}{5}$ and $\text{Var}(Y_1) = \frac{\text{Var}(X)}{200} = 2$, but this is actually not required for the solution.

3. (a) (5 points) A fair 6-sided die is rolled once. Find the expected number of additional rolls needed to obtain a value at least as large as that of the first roll, given that the first roll is 3.

Solution: Given that the first roll is 3, we will roll a die until we have 3, 4, 5 or 6. Probability of it is $\frac{2}{3}$, so the distribution of number of rolls is $\text{Geom}(\frac{2}{3})$ and expectation is $\frac{3}{2}$.

- (b) (6 points) A fair 6-sided die is rolled once. Find the expected number of additional rolls needed to obtain a value at least as large as that of the first roll.

Solution: Let N the number of additional rolls and X_1 be the number on the first roll. Now

$$E[N] = P(X_1 = 1)E[N|X_1 = 1] + P(X_1 = 2)E[N|X_1 = 2] + P(X_1 = 3)E[N|X_1 = 3] \\ + P(X_1 = 4)E[N|X_1 = 4] + P(X_1 = 5)E[N|X_1 = 5] + P(X_1 = 6)E[N|X_1 = 6].$$

$P(X_1 = i) = \frac{1}{6}$, and, similarly to part (a), we have $N|X_1 = i \sim \text{Geom}(\frac{6-i+1}{6})$, and $E[N|X_1 = i] = \frac{6}{6-i+1}$. We can write $E[N|X_1] = \frac{6}{7-X_1}$ or do it explicitly in each case:

$$\begin{aligned} N|X_1 = 1 &\sim \text{Geom}(\frac{6}{6}), & E[N|X_1 = 1] &= \frac{6}{6}, \\ N|X_1 = 2 &\sim \text{Geom}(\frac{5}{6}), & E[N|X_1 = 2] &= \frac{6}{5}, \\ N|X_1 = 3 &\sim \text{Geom}(\frac{4}{6}), & E[N|X_1 = 3] &= \frac{6}{4}, \\ N|X_1 = 4 &\sim \text{Geom}(\frac{3}{6}), & E[N|X_1 = 4] &= \frac{6}{3}, \\ N|X_1 = 5 &\sim \text{Geom}(\frac{2}{6}), & E[N|X_1 = 5] &= \frac{6}{2}, \\ N|X_1 = 6 &\sim \text{Geom}(\frac{1}{6}), & E[N|X_1 = 6] &= \frac{6}{1}. \end{aligned}$$

Combining everything in the formula:

$$E[N] = \frac{1}{6} \cdot \frac{6}{6} + \frac{1}{6} \cdot \frac{6}{5} + \frac{1}{6} \cdot \frac{6}{4} + \frac{1}{6} \cdot \frac{6}{3} + \frac{1}{6} \cdot \frac{6}{2} + \frac{1}{6} \cdot \frac{6}{1} = \frac{1}{6} + \frac{1}{5} + \frac{1}{4} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1} = \frac{147}{60} = 2 \frac{9}{20}.$$

4. We are given a biased coin and we are told that because of manufacturing defects, the probability of heads, denoted by Y , is itself random, with a distribution over the interval $[0, 0.5]$ given by density $1 + 4y$. We toss the coin a fixed number n of times, and we let X be the number of heads obtained.

(a) (6 points) Find $E[X|Y]$.

Solution: Distribution of X given $Y = y$ is $\text{Bin}(n, y)$, so $E[X|Y = y] = ny$ and $E[X|Y] = nY$.

(b) (6 points) Find $E[X]$.

Solution:

$$E[X] = E[E[X|Y]] = E[nY] = nE[Y] = n \int_0^{0.5} y(1 + 4y)dy = n \left(\frac{y^2}{2} + \frac{4y^3}{3} \right) \bigg|_0^{0.5} = \boxed{\frac{7}{24}n}.$$

5. (8 points) Suppose $X \sim \text{Exp}(\lambda)$, that is, X is an exponential random variable with parameter $\lambda > 0$. Find the probability density function of X^2 .

Solution: The p.d.f. of X is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = X^2$ and note that Y is a non-negative random variable. Hence for any $y \geq 0$, the c.d.f. of Y is

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^2 \leq y) \\ &= P(X \leq \sqrt{y}) \quad X \text{ is non-negative, and so } X \geq -\sqrt{y} \text{ is automatic.} \\ &= F_X(\sqrt{y}) \\ &= \int_{-\infty}^{\sqrt{y}} f_X(x) dx = \int_0^{\sqrt{y}} \lambda e^{-\lambda x} \\ &= 1 - e^{-\lambda\sqrt{y}}. \end{aligned}$$

Taking derivative of $F_Y(y)$, we obtain for $y \geq 0$,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{\lambda e^{-\lambda\sqrt{y}}}{2\sqrt{y}}.$$

Therefore, recalling that Y is non-negative, the p.d.f. of Y is

$$f_Y(y) = \begin{cases} \frac{\lambda e^{-\lambda\sqrt{y}}}{2\sqrt{y}} & \text{if } y > 0, \\ 0 & \text{otherwise.} \end{cases}$$

6. An urn contains six balls, one is red, three are green, two are yellow. Randomly draw three balls from the urn **without** replacement, and let R and G denote the number of red and green balls (within the three balls), respectively.
- (a) (6 points) Compute $P(R = 1, G = 1)$.

Solution: Let X_1, X_2, X_3 denote the colors of the first three balls. Then X_1, X_2, X_3 are exchangeable. Let r,g,y denote the colors red, green and yellow, respectively. Let Y denote the number of yellow balls. Then

$$\begin{aligned}
 P(R = 1, G = 1) &= P(R = 1, G = 1, Y = 1) \\
 &= P(\{X_1, X_2, X_3\} = \{r, g, y\}) \\
 &= 3!P(X_1 = r, X_2 = g, X_3 = y) \quad \text{Here we used exchangeability} \\
 &= 6 \times \frac{1}{6} \times \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}.
 \end{aligned}$$

- (b) (6 points) Compute the expectations $E[R]$ and $E[G]$.

Solution: For $i = 1, 2, 3$, we let

$$I_i = \begin{cases} 1 & \text{if the } i\text{-th ball is red} \\ 0 & \text{otherwise,} \end{cases} \quad J_i = \begin{cases} 1 & \text{if the } i\text{-th ball is green} \\ 0 & \text{otherwise.} \end{cases}$$

Then $R = I_1 + I_2 + I_3$ and $G = J_1 + J_2 + J_3$. Note that I_i 's are exchangeable, and so are the J_i 's. Moreover,

$$E[I_1] = P(\text{the first ball is red}) = \frac{1}{6}, \quad E[J_1] = P(\text{the first ball is green}) = \frac{3}{6} = \frac{1}{2}.$$

Therefore, by the linearity of expectations,

$$E[R] = E[I_1] + E[I_2] + E[I_3] = 3E[I_1] = \frac{1}{2}, \quad E[G] = E[J_1] + E[J_2] + E[J_3] = 3E[J_1] = \frac{3}{2}$$

(c) (8 points) Compute the covariance $\text{Cov}(R, G)$.

Solution:

Method 1.

Since $\text{Cov}(R, G) = E[RG] - E[R]E[G]$, it remains to compute $E[RG]$ (given that $E[R], E[G]$ are obtained in (b)). Note that

$$\begin{aligned} E[RG] &= 0 + 1 \times 1 \times P(R = 1, G = 1) + 1 \times 2P(R = 1, G = 2) \\ &\stackrel{\text{by (a)}}{=} \frac{3}{10} + 2P(R = 1, G = 2). \end{aligned}$$

Moreover, using the knowledge of counting,

$$P(R = 1, G = 2) = \frac{1 \times \binom{3}{2}}{\binom{6}{3}} = \frac{3}{20}.$$

Hence $E[RG] = \frac{3}{10} + 2 \times \frac{3}{20} = \frac{3}{5}$ and so

$$\text{Cov}(R, G) = E[RG] - E[R]E[G] = \frac{3}{5} - \frac{1}{2} \times \frac{3}{2} = -\frac{3}{20}.$$

Method 2

$$\begin{aligned} \text{Cov}(R, G) &= \sum_{i,j=1}^3 \text{Cov}(I_i, J_j) \\ &= \sum_{i=1}^3 \text{Cov}(I_i, J_i) + \sum_{i \neq j} \text{Cov}(I_i, J_j) \\ &= 3\text{Cov}(I_1, J_1) + 6\text{Cov}(I_1, J_2), \end{aligned}$$

where we used exchangeability in the last step. Since

$$\text{Cov}(I_1, J_1) = E[I_1 J_1] - E[I_1]E[J_1] = 0 - \frac{1}{6} \cdot \frac{1}{2} = -\frac{1}{12}$$

and

$$\text{Cov}(I_1, J_2) = E[I_1 J_2] - E[I_1]E[J_2] = \frac{1}{6} \cdot \frac{3}{5} - \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{60},$$

we conclude that

$$\text{Cov}(R, G) = 3 \cdot \left(-\frac{1}{12}\right) + 6 \cdot \frac{1}{60} = -\frac{3}{20}.$$

7. (8 points) Let X and Y be independent random variables with distributions $\text{Unif}(0, 2)$ and $\text{Unif}(1, 3)$. Let $W = X + Y$. Find the probability density function of W .

Solution: Probability densities of X and Y are

$$f_X(x) = \begin{cases} \frac{1}{2} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{2} & \text{if } 1 < y < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Density of $W = X + Y$ is the convolution

$$f_w(w) = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x) dx.$$

The product $f_X(x)f_Y(w - x)$ is nonzero and equal to $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ if and only if $0 \leq x \leq 2$ and $1 \leq w - x \leq 3$. The second inequality is equivalent to $w - 3 \leq x \leq w - 1$. To simultaneously satisfy $0 \leq x \leq 2$ and $w - 3 \leq x \leq w - 1$, we take the larger of the lower bounds and the smaller of the upper bounds on x . Thus $f_X(x)f_Y(w - x) = \frac{1}{4}$ if and only if

$$\max(0, w - 3) \leq x \leq \min(2, w - 1). \quad (1)$$

If $w - 1 < 0$ or $w - 3 > 2$, then $\min(2, w - 1) < \max(0, w - 3)$, so there is no x satisfying (1). Thus $f_W(w) = 0$ if $w < 0$ or $w > 5$. For $1 \leq w \leq 5$ we have

$$f_W(w) = \int_{\max(0, w-3)}^{\min(2, w-1)} \frac{1}{4} dx = \frac{1}{4}(\min(2, w - 1) - \max(0, w - 3)).$$

Note that $\min(2, w - 1) = 2$ and $\max(0, w - 3) = w - 3$ if $w \geq 3$, and that $\min(2, w - 1) = w - 1$ and $\max(0, w - 3) = 0$ if $w \leq 3$. Thus we have

$$f_W(w) = \begin{cases} \frac{1}{4}(w - 1) & \text{if } 1 \leq w \leq 3 \\ \frac{1}{4}(5 - w) & \text{if } 3 < w \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Table of Named Distributions

Name	p.m.f. or p.d.f.	$E[X]$	$Var(X)$
Ber(p)	$p_X(0) = 1 - p, p_X(1) = p$	p	$p(1 - p)$
Bin(n, p)	$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, 0 \leq k \leq n$	np	$np(1 - p)$
Geom(p)	$p_X(k) = p(1 - p)^{k-1}, k \geq 1$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
NegBin(k, p)	$p_X(n) = \binom{n-1}{k-1} p^k (1 - p)^{n-k}, n \geq k$	$\frac{k}{p}$	$\frac{k(1 - p)}{p^2}$
HyperGeom(N, N_A, n)	$p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}$	$\frac{nN_A}{N}$	$\frac{nN_A(N - N_A)(N - n)}{N^2(N - 1)}$
Poisson(λ)	$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, k \geq 0$	λ	λ
Unif(a, b)	$f_X(x) = \frac{1}{b - a}, x \in [a, b]$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
N(μ, σ^2)	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$	μ	σ^2
Exp(λ)	$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(n, λ)	$f_X(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n - 1)!}, x \geq 0$	$\frac{n}{\lambda}$	$\frac{n}{\lambda^2}$

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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1. (12 points) For this problem, you need to provide **answers only**.

(a) Let X be the random variable with moment generating function

$$M_X(t) = \frac{1}{4} + \frac{1}{8}e^{2t} + \frac{1}{8}e^{3t} + \frac{1}{2}e^{5t}.$$

What is the p.m.f. of X ?

Solution:

$$P(X = 0) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{8}, \quad P(X = 3) = \frac{1}{8}, \quad P(X = 5) = \frac{1}{2}.$$

The rest are 0.

(b) A full house in poker consists of 5 cards where three are the same value and the remaining two are the same value (e.g. three Q's and two 4's). If 5 cards are dealt from a shuffled deck without replacement, what is the probability of drawing a full house?

Solution: There are

$$\begin{aligned} &\binom{13}{1} \text{ ways of selecting the value for the 3-of-a-kind} \\ &\binom{4}{3} \text{ ways of selecting those 3 of the 4 cards} \\ &\binom{12}{1} \text{ ways of selecting the value for the pair} \\ &\binom{4}{2} \text{ ways of selecting the pair from the 4 cards of the value.} \end{aligned}$$

Hence

$$p = \frac{\binom{13}{1} \binom{4}{3} \binom{12}{1} \binom{4}{2}}{\binom{52}{5}}.$$

(c) Let X and Y be i.i.d. $\text{Unif}[0, 1]$. Compute the covariance of $X + Y$ and $X - Y$.

Solution:

$$\begin{aligned} \text{Cov}(X + Y, X - Y) &= \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y) \\ &= \text{Var}(X) - 0 + 0 - \text{Var}(Y) = 0. \end{aligned}$$

(d) We flip a coin over and over again. Let N_n be the number of flips required to see the n th Heads. What is $\lim_{n \rightarrow \infty} P(N_n > 3n)$?

Solution: $N_n = \sum_{j=1}^n X_j$ where X_j are i.i.d. $\text{Geom}(1/2)$. So $E[X_j] = 2$ and

$$P(N_n/n > 3) \leq P\left(\left|\frac{N_n}{n} - 2\right| > 1\right) \rightarrow 0.$$

2. (15 points) **Please provide explanation for all parts.**

Suppose that X and Y are random variables with

$$E[X] = 2, \quad E[X^2] = 5, \quad E[Y] = 4, \quad \text{Cov}(X, Y) = -4.$$

(a) What is $\text{Var}(X)$.

Solution: $\text{Var}(X) = E[X^2] - E[X]^2 = 5 - 4 = 1.$

(b) What is $E[XY]$?

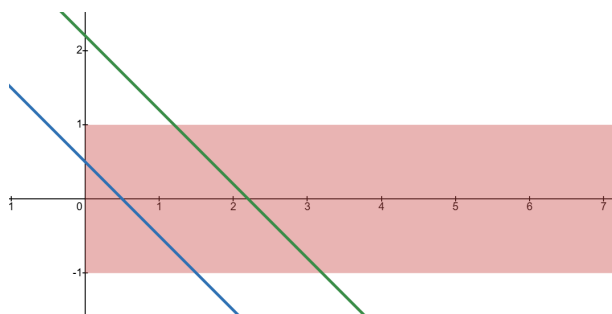
Solution: $E[XY] = \text{Cov}(X, Y) + E[X]E[Y] = -4 + 8 = 4.$

(c) What is $\text{Cov}(2X + Y, X)$?

Solution: $\text{Cov}(2X + Y, X) = 2 \text{Var}(X) + \text{Cov}(Y, X) = 2 \cdot 1 + (-4) = -2.$

3. (8 points) Let X and Y be two independent random variables where $X \sim \text{Exp}(1)$ and $Y \sim \text{Unif}(-1, 1)$. What is the probability density function of $Z = X + Y$?

Solution: For $z \in (-1, 1)$ we integrate the convolution from $x = 0$ to $x = z$, while for $z \geq 1$ we integrate from $x = z - 1$ to $x = z$. See the picture below.



For $z \in (-1, 1)$:

$$f_Z(z) = \int_0^{z+1} f_X(x) f_Y(z-x) dx = \int_0^{z+1} e^{-x} \cdot \frac{1}{2} dx = -\frac{1}{2} e^{-x} \Big|_{x=0}^{z+1} = \frac{1}{2} (1 - e^{-z-1}).$$

For $z \geq 1$

$$f_Z(z) = \int_{z-1}^{z+1} e^{-x} \cdot \frac{1}{2} dx = -\frac{1}{2} e^{-x} \Big|_{x=z-1}^{z+1} = \frac{1}{2} (e^{-(z-1)} - e^{-(z+1)})$$

For $z \leq -1$, $f_Z(z) = 0$. Hence

$$f_Z(z) = \begin{cases} 0 & : z \leq -1 \\ \frac{1}{2}(1 - e^{-z-1}) & : z \in (-1, 1) \\ \frac{1}{2}(e^{-(z-1)} - e^{-(z+1)}) & : z \geq 1. \end{cases}$$

4. Let X and Y be jointly continuous random variables with joint density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{y}e^{-2y} & : 0 < x < y \\ 0 & : \text{else} \end{cases}$$

- (a) (5 points) Find the marginal density of Y , i.e. $f_Y(y)$. Specify the name of the distribution and its parameters.

Solution: For $y > 0$:

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_0^y \frac{2}{y}e^{-2y} dx = 2e^{-2y}.$$

For $y \leq 0$ the p.d.f. of Y is zero. So $Y \sim \text{Exp}(2)$.

- (b) (5 points) Determine the conditional distribution $f_{X|Y}(x|y)$ for each $y > 0$?

Solution: For $x \in (0, y)$:

$$f_{X|Y}(x|y) = \frac{\frac{2}{y}e^{-2y}}{2e^{-2y}} = \frac{1}{y}$$

and for $x \geq y$ the conditional p.d.f. is 0. So, for all $y > 0$:

$$f_{X|Y}(x|y) = \begin{cases} y^{-1} & : x \in (0, y) \\ 0 & : \text{else} \end{cases}$$

and, actually, $X|Y = y \sim \text{Unif}[0, y]$.

Problem 4, cont.

(c) (5 points) What is $E[X|Y = y]$ for each $y > 0$?

Solution:

$$E[X|Y = y] = E[\text{Unif}[0, y]] = \frac{y}{2}.$$

(d) (5 points) What is $E[X]$?

Solution: From (c) $E[X|Y] = \frac{1}{2}Y$. So

$$E[X] = E[E[X|Y]] = E[\tfrac{1}{2}Y] = \tfrac{1}{2}E[\text{Exp}(2)] = \tfrac{1}{4}.$$

5. There is an urn with 30 red balls, 20 green balls, and 20 yellow balls. You sample 40 balls from the urn without replacement, one after the other.
- (a) (5 points) Given that the last ball drawn is red, what is the probability the first ball drawn is yellow?

Solution: Let D_k be the color of the k th draw and let the colors be $\{\mathbf{R}, \mathbf{G}, \mathbf{Y}\}$ for red, green and yellow. Then D_k are exchangeable and

$$\begin{aligned} P(D_1 = \mathbf{Y} | D_{40} = \mathbf{R}) &= \frac{P(D_1 = \mathbf{Y}, D_{40} = \mathbf{R})}{P(D_{40} = \mathbf{R})} \\ &= \frac{P(D_2 = \mathbf{Y}, D_1 = \mathbf{R})}{P(D_1 = \mathbf{R})} \\ &= P(D_2 = \mathbf{Y} | D_1 = \mathbf{R}) = \frac{20}{69}. \end{aligned}$$

- (b) (6 points) Let X be the number of times that you draw three consecutive balls of the same color. What is $E[X]$?

Solution: Let I_k be the indicator that balls $k-2$, $k-1$, k are the same color, i.e.

$$I_k = \begin{cases} 1 & : D_{k-2} = D_{k-1} = D_k \\ 0 & : \text{else} \end{cases} \quad k = 3, 4, \dots, 40.$$

There are 38 indicators and, by exchangeability of D_k

$$E[I_k] = E[I_3] = P(D_3 = D_2 = D_1).$$

This probability is

$$\begin{aligned} &P(D_3 = D_2 = D_1 = \mathbf{R}) + P(D_3 = D_2 = D_1 = \mathbf{G}) + P(D_3 = D_2 = D_1 = \mathbf{Y}) \\ &= \frac{30 \cdot 29 \cdot 28}{70 \cdot 69 \cdot 68} + \frac{20 \cdot 19 \cdot 18}{70 \cdot 69 \cdot 68} + \frac{20 \cdot 19 \cdot 18}{70 \cdot 69 \cdot 68} \end{aligned}$$

So

$$E[X] = 38E[I_3] = 38 \left(\frac{30 \cdot 29 \cdot 28}{70 \cdot 69 \cdot 68} + 2 \cdot \frac{20 \cdot 19 \cdot 18}{70 \cdot 69 \cdot 68} \right)$$

6. Let X be a non-negative random variable. Each part requires an explanation.

(a) (5 points) We know that $E[X] = 200$, what can we say about $P(X > 300)$?

Solution:

$$P(X \geq 300) \leq \frac{E[X]}{300} = \frac{2}{3} = 0.6667.$$

(b) (5 points) Suppose we also know that $\text{Var}(X) = 3600$, what can we say about $P(X > 300)$?

Solution: Chebyshev's inequality says that $P(|X - E[X]| > c) \leq \text{Var}(X)/c^2$ for $c > 0$. Hence

$$\begin{aligned} P(X \geq 300) &= P(X - 200 \geq 100) \leq P(|X - 200| \geq 100) \\ &\leq \text{Var}(X)/100^2 = \frac{3600}{10000} = \frac{36}{1000} = 0.036. \end{aligned}$$

(c) (5 points) Suppose we also know that X is the sum of 50 i.i.d. random variables. What is the expectation and variance of the summand? That is if $X = \sum_{j=1}^{50} Y_j$, where Y_j are i.i.d., then what is $E[Y_j]$ and $\text{Var}(Y_j)$?

Solution: Suppose $X = Y_1 + \dots + Y_{50}$. We have $E[Y_i] = 200/50 = 4$, $\text{Var}(Y_i) = 3600/50 = 72$.

(d) (5 points) Given the information in (c) provide an estimate for the precise value of $P(X > 300)$, assuming it is reasonable.

Solution: We can estimate $P(X > 400)$ by the CLT:

$$\begin{aligned} P(X > 300) &= P\left(\frac{Y_1 + Y_2 + \dots + Y_{50} - 200}{\sqrt{3600}} > \frac{300 - 200}{60}\right) \approx 1 - \Phi(1.67) \approx 0.0475 \\ 1 - \Phi(1.66) &\approx 0.0485 \end{aligned}$$

7. A fair 6-sided die is rolled, and then a fair coin is flipped as many times as the die roll says, e.g., if the result of the die roll is a 3, then the coin is flipped 3 times. Let X be the result of the die roll and Y be the number of times the coin lands Heads.

(a) (5 points) Find the joint pmf of X and Y .

Solution:

$$P(X = k, Y = \ell) = \frac{1}{6} \binom{k}{\ell} \frac{1}{2^k}, \quad k \in \{1, 2, 3, 4, 5, 6\}, \quad \ell \in \{0, 1, 2, \dots, k\}.$$

(b) (5 points) Find $P(Y = 5)$.

Solution:

$$P(Y = 5) = P(Y = 5, X = 5) + P(Y = 5, X = 6) = \frac{1}{6} \binom{5}{5} \frac{1}{32} + \frac{1}{6} \binom{6}{5} \frac{1}{64} = \frac{1}{48}.$$

(c) (5 points) What is $E[X|Y = 5]$?

Solution: We use part (c) and Bayes' formula.

$$\begin{aligned} E[X|Y = 5] &= 5P(X = 5|Y = 5) + 6P(X = 6|Y = 5) \\ &= 5 \frac{P(X = 5, Y = 5)}{P(Y = 5)} + 6 \frac{P(X = 6, Y = 5)}{P(Y = 5)} \\ &= 5 \frac{\frac{1}{6} \binom{5}{5} \frac{1}{32}}{\frac{1}{6} \binom{5}{5} \frac{1}{32} + \frac{1}{6} \binom{6}{5} \frac{1}{64}} + 6 \frac{\frac{1}{6} \binom{6}{5} \frac{1}{64}}{\frac{1}{6} \binom{5}{5} \frac{1}{32} + \frac{1}{6} \binom{6}{5} \frac{1}{64}} \\ &= 5 \frac{1}{4} + 6 \frac{3}{4} = 5.75. \end{aligned}$$

8. We have two bins. The first bin has 6 blue marbles and 4 yellow marbles. The second bin has 3 blue marbles and 4 yellow marbles. We flip a coin. If it is heads, we sample from the first bin, otherwise we sample from the second.
- (a) (4 points) If the marble we select is yellow, what is the probability that we chose the first bin?

Solution: Let A_i be the event that bin i was chosen ($i = 1, 2$) and Y_j the event that draw j ($j = 1, 2$) is yellow.

$$\begin{aligned} P(A_1|Y_1) &= \frac{P(Y_1|A_1)P(A_1)}{P(Y_1|A_1)P(A_1) + P(Y_1|A_2)P(A_2)} \\ &= \frac{\frac{4}{10} \cdot \frac{1}{2}}{\frac{4}{10} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{1}{2}} = \frac{14}{34} \approx 0.4118. \end{aligned}$$

- (b) (4 points) Now suppose we put the yellow marble from (a) back in the bin it was drawn from and then draw a marble from the same bin. This marble is also yellow. What is the probability we chose the first bin?

Solution: This question asks for the conditional probability of A_1 , given that two draws with replacement from the chosen urn yield yellow. We assume that draws with replacement from the same urn are independent. This translates into conditional independence of Y_1 and Y_2 , given A_i .

$$\begin{aligned} P(A_1|Y_1Y_2) &= \frac{P(Y_1Y_2|A_1)P(A_1)}{P(Y_1Y_2|A_1)P(A_1) + P(Y_1Y_2|A_2)P(A_2)} \\ &= \frac{P(Y_1|A_1)P(Y_2|A_1)P(A_1)}{P(Y_1|A_1)P(Y_2|A_1)P(A_1) + P(Y_1|A_2)P(Y_2|A_2)P(A_2)} \\ &= \frac{\frac{4}{10} \cdot \frac{4}{10} \cdot \frac{1}{2}}{\frac{4}{10} \cdot \frac{4}{10} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{4}{7} \cdot \frac{1}{2}} = \frac{196}{596} \approx 0.3289. \end{aligned}$$

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Table of Named Distributions

Name	p.m.f. or p.d.f.	$E[X]$	$Var(X)$
Ber(p)	$p_X(0) = 1 - p, p_X(1) = p$	p	$p(1 - p)$
Bin(n, p)	$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, 0 \leq k \leq n$	np	$np(1 - p)$
Geom(p)	$p_X(k) = p(1 - p)^{k-1}, k \geq 1$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
NegBin(k, p)	$p_X(n) = \binom{n-1}{k-1} p^k (1 - p)^{n-k}, n \geq k$	$\frac{k}{p}$	$\frac{k(1 - p)}{p^2}$
HyperGeom(N, N_A, n)	$p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}$	$\frac{nN_A}{N}$	$\frac{nN_A(N - N_A)(N - n)}{N^2(N - 1)}$
Poisson(λ)	$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, k \geq 0$	λ	λ
Unif(a, b)	$f_X(x) = \frac{1}{b - a}, x \in [a, b]$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
N(μ, σ^2)	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} 2e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exp(λ)	$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(n, λ)	$f_X(x) = \frac{\lambda^n x^{n-1} e^{-\lambda}}{(n - 1)!}, x \geq 0$	$\frac{n}{\lambda}$	$\frac{n}{\lambda^2}$

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998